MATH 207 Discrete Mathematics

Fall 2023

Instructor: Nicholas Owad
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Class time: TuTh 9:50 – 11:15
Classroom: Hodson Science Center 316
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Rationale for the course

Course description

An introduction to basic concepts and techniques of discrete mathematics. Topics include logic, sets, positional numeration systems, mathematical induction, elementary combinatorics, algorithms, matrices, recursion and the basic concepts of graphs and trees. The relationship to the computer will be stressed throughout.

Prerequisite

MATH 120 (Precalculus) or Level III placement on the Basic Math Skills Inventory or permission of the instructor.

Course goals and learning outcomes

Professors decide a few major goals or important learning outcomes in each course. I will be thinking about the goals below throughout the semester as I plan class activities, write homework assignments, and create midterms and final exams.

Course-specific goals

At the end of MATH 207, you should be able to:

1. Demonstrate skill in problem solving by applying critical reasoning and quantitative methods to solve mathematical problems.
2. Construct mathematical arguments and write mathematical proofs using sound logic, correct use of terminology and notation, and clear communication.
3. Demonstrate skill in written and oral communication of mathematics by
    a. Speaking about mathematics coherently and logically with mastery of disciplinary style and conventions.
    b. Writing about mathematics coherently and logically with mastery of disciplinary style and conventions.

Learning outcomes

In addition to the mathematics program goals, here are several course-specific learning goals. Upon successful completion of this course, you will be able to:

1. formulate and interpret statements presented in Boolean logic; reformulate statements from com- mon language to formal logic and apply truth tables to them

2. evaluate elementary mathematical arguments and identify fallacious reasoning
3. use mathematical and logical notation to define and formally reason about mathematical concepts such as sets, relations, and functions, and discrete structures like trees and graphs
4. write basic proofs for facts in elementary number theory
5. construct an inductive hypothesis and carry out simple induction proofs
6. apply counting arguments to basic problems
7. solve a variety of recurrence relations

Format and procedures

MATH 207 will be held in-person, meeting two times per week during our scheduled 1-hour-and-25-minute block in our assigned room. We will not record or stream classes unless otherwise noted, so attendance is required to complete the material.

Course expectations and requirements

Texts

We will use *Discrete Mathematics – An Open Introduction*, 3rd Edition, by Oscar Levin. You do not have to buy this textbook unless you want a physical copy. It is available for free online at http://discrete.openmathbooks.org/dmoi3.html.

Reading mathematics is a challenging endeavor to undertake. You must practice this skill. I expect you to put effort into not just reading the words but understanding what the author is attempting to relay. Be sure to read and re-read the content until you have some grasp of the material.

Technology

Calculating tools

Many tools are available for calculating but most are not suited for use in Discrete mathematics. You are always welcome to use the following.

1. Desmos at https://desmos.com. Desmos is a free online graphing tool. Graphs can be created on Desmos also at https://www.desmos.com/geometry.
2. Google has a built in calculator which you can access with this link. Or you can search the term 'calculator' in the standard google search bar. Alternatively, you can just enter your expression in the search bar.

Writing tools

Writing is an important part of how you will learn mathematics this semester and improve your quantitative literacy. You may use Google docs, Microsoft Word (only the desktop version has the equation editor), or if you are ambitious, LaTeX to type mathematics.

Homework

Learning mathematics is like learning a language in that it takes a lot of regular practice. To help you practice, I will assign homework. You will find all homework assignments in the "Assignments" section of our Blackboard page, and you will turn in your completed homework to me in person at the beginning of class on Thursdays. You should check Blackboard regularly for these assignments, even if I do not specifically announce that I will post an assignment.

You may collaborate on homework assignments with other students, but you must write up the work that you turn in on your own. What you turn in should reflect your own personal understanding of the material. This will help me guide your learning. If I find homework submissions that are practically identical, I will not accept them for credit.

Midterms and final exam

We will have two midterm exams and a final exam. As with any class meeting, you will be allowed to work in person or remotely on an exam day. I may choose to make any of these exams remote work for everyone in the class.

I have tentatively scheduled the first midterm exam for **Thursday, September 28th**, the second midterm exam for **Thursday, November 9th**, and the Final exam will be scheduled during **December 6th – 8th or 11th – 12th.** As soon as Final Exam times are announced, I will let you know when our Final Exam is scheduled.

Before each midterm exam, I will tell you which topics will be included on the test. Everything we have covered in the class up to that point is possible to show up on the midterm, but usually it will focus on the new material. The final exam will be comprehensive, including topics from the entire course.

Academic Honor Code and the Code of Conduct

A complete description of Hood's Academic Honor code is available in Appendix B of the Student Handbook which is available at http://www.hood.edu/policies/ . This page also has information about Hood's Code of Conduct as well as a statement of *Policy 55: Prevention and Resolution of Bullying, Discrimination, and Harassment at Hood College.*

Grading procedures

I will post all grades for assignments in the gradebook on our Blackboard page. I will do my best to give you feedback quickly on anything that you turn in. I will compute your final course grade according to the following breakdown, but these are also tentative.

Course part	**Points**
Homework	200
Group Work	200
Midterm 1	100
Midterm 2	100
Final exam	200
Total	**800**

Accessibility

If you have a disability or a personal circumstance that will affect your learning in this course, please let me know as soon as possible so that we can discuss the best ways to meet your needs.

Students who need accommodation for disabilities are strongly encouraged to contact the Office of Accessibility Services by e-mail at AccessibilityServices@hood.edu or by phone at 301-696-3421 to obtain official letters of accommodation for all their courses.

Tutoring and Student Success

Free tutoring, both online and in-person, is available through our Student Success Office. More information about this and other support services is available here: https://www.hood.edu/academics/josephine-steiner-student-success-center

Academic Honor Code and the Code of Conduct

As always, you are expected to follow all policies as described in the Hood College Student Policy Documents available at http://www.hood.edu/policies/.

A complete description of Hood's Academic Honor code is available in Appendix B of the Student Handbook. All work you turn in should be your own (for team projects, it should be the work of the team). As already mentioned, collaboration is encouraged on homework, labs, and projects, but must be noted. If you have any questions about what is acceptable, please do not hesitate to ask. Asking *before* completing the assignment prevents unintentional violations of the honor code.

Inclusivity Statement

Every student in this class, regardless of background, sex, gender, race, ethnicity, class, political affiliation, physical or mental ability or any identity category, is a valued and equal member of the group. We all bring different experiences to this class, and no one experience has more value than another. In fact, it is our different experiences that will enrich the course content. Furthermore, in this classroom, you have the right to determine your own identity. You have the right to be called by whatever name you wish, and for that name to be pronounced correctly. You have the right to be referred to by whatever pronoun you identify. You have the right to adjust those things at any point. If there are aspects of the instruction of this course that result in barriers to your inclusion or a sense of alienation from the course content, please contact me privately without fear of reprisal. If you feel uncomfortable contacting me, please contact the Office of the Dean of Students. More information and resources are also available to you here at this link https://www.hood.edu/hood-community/diversity-inclusion